Technical Report - Human Resources Candidate Ranking

Duplicate-safe lexical ranking, fold-safe model selection, and recruiter feedback

Selected Stage-1 model: normalized hybrid retrieval (BM25 + word TF-IDF + character TF-IDF) with lexical overlap and HR/intent features, Ridge alpha = 10. Mean grouped-CV NDCG@10 = 0.9628 +/- 0.0621; MAP@10 = 1.000; MRR = 1.000.

1. Objective and data

The project ranks candidate profiles for two HR-oriented search intents: "aspiring human resources" and "seeking human resources". Scores are relative relevance scores used for ordering; they are not hiring probabilities or predictions of employee performance.

Item	Value / treatment
Source rows	104
Unique duplicate-safe profile groups	53
Primary text field	job_title
Human relevance scale	0-3
Primary metric	NDCG@10
MAP/MRR relevant threshold	grade >= 2

Notebook 01 checks that the original five source fields match the labelled file, IDs are unique, all grades are complete and within 0-3, duplicate groups carry one consistent grade, and the final grouping produces exactly 53 unique profiles.

Relevance labels

The unique-profile distribution is 17 / 2 / 28 / 6 for grades 0 / 1 / 2 / 3. After propagation to all source rows, the distribution is 27 / 10 / 47 / 20. Duplicate rows remain traceable but do not create additional profile observations.

2. Fold-safe model selection

Five-fold GroupKFold is applied at the 53-profile level. Duplicate copies of the same profile cannot appear in both training and validation.

- Word TF-IDF vocabulary and IDF are fitted from training titles only.
- Character TF-IDF vocabulary and IDF are fitted from training titles only.
- BM25 document-frequency and average-document-length statistics are learned from training titles only.
- Validation titles and the two search queries are transformed or scored with the training-fitted representation.
- Ridge is trained on the resulting query-profile feature rows and evaluated on held-out profile groups.

The broad search contains 3 preprocessing modes x 4 retrieval representations x 4 feature bundles x 4 Ridge penalties = 192 configurations. It includes surface, normalized, and Porter2 preprocessing; word TF-IDF, word+character TF-IDF, BM25, and hybrid representations; retrieval-only through retrieval+overlap+intent+connections feature bundles; and alpha values 0.1, 1, 10, and 100.

3. Model-selection results

Configuration	Mean NDCG@10	SD	MAP@10	MRR
Selected: normalized hybrid + overlap/intent, alpha 10	0.9628	0.0621	1.000	1.000
Normalized word TF-IDF + overlap, alpha 10	0.9622	0.0634	1.000	1.000
Normalized word+char TF-IDF + overlap, alpha 10	0.9622	0.0634	1.000	1.000
Best with connection count	0.9562	0.0649	1.000	1.000

The selected model is the highest-mean NDCG@10 configuration in the cleaned 192-run screen. Connection count is excluded because the best connection-inclusive variant scores lower.

Direct lexical baselines

Baseline	Mean NDCG@10	SD	MAP@10	MRR
Character TF-IDF	0.9535	0.0582	0.9938	1.000
BM25	0.9527	0.0577	0.9909	1.000
Word TF-IDF	0.9359	0.0668	0.9874	1.000

BM25 versus word TF-IDF

Across the ten matched observations (5 folds x 2 queries), BM25 minus word TF-IDF has mean NDCG@10 difference +0.0168, paired t-test p = 0.278, and 95% CI [-0.0161, +0.0498]. Win/tie/loss is 3/6/1. The result does not establish a clear superiority claim.

4. Final Stage-1 ranking

After model selection, the chosen Ridge configuration is fitted on all 53 unique development profiles. Final TF-IDF representations are fitted on the candidate titles only; queries are transformed afterward. Each profile receives the maximum normalized prediction across the two search intents.

Rank	ID	Title	Stage-1 score
1	99	Seeking Human Resources Position	1.0000
2	3	Aspiring Human Resources Professional	0.9770
3	97	Aspiring Human Resources Professional	0.9770
4	6	Aspiring Human Resources Specialist	0.9727
5	28	Seeking Human Resources Opportunities	0.9576
6	73	Aspiring HR Manager, seeking internship in HR	0.8996
7	10	Seeking HR HRIS and Generalist Positions	0.8482
8	27	Aspiring HR Management student seeking internship	0.8390
9	79	Liberal Arts Major. Aspiring HR Analyst.	0.8301
10	72	Business Management Major and Aspiring HR Manager	0.8133

5. Stage-2 recruiter feedback

The demonstration maps source ID 3 to its duplicate-safe profile group, uses it as positive feedback, uses no rejected profile, and applies Rocchio parameters alpha = 1.0, beta = 0.75, gamma = 0.15. The feedback TF-IDF space is fitted on candidate titles only. Stage 2 blends 65% Stage 1 with 35% feedback.

Stage-2 rank	ID	Stage-1 rank	Change	Title
1	3	2	+1	Aspiring Human Resources Professional
2	97	3	+1	Aspiring Human Resources Professional
3	99	1	-2	Seeking Human Resources Position
4	6	4	0	Aspiring Human Resources Specialist
5	28	5	0	Seeking Human Resources Opportunities
6	74	11	+5	Human Resources Professional
7	73	6	-1	Aspiring HR Manager, seeking internship in HR
8	66	12	+4	Experienced Retail Manager and aspiring HR Professional
9	10	7	-2	Seeking HR HRIS and Generalist Positions
10	27	8	-2	Aspiring HR Management student seeking internship

The largest upward movement in the demonstration is ID 76, from Stage-1 rank 28 to Stage-2 rank 20 (+8). ID 1 rises from 27 to 21 (+6), and ID 74 rises from 11 to 6 (+5). This shows controlled personalization rather than replacement of the base ranking.

6. Interpretation and limitations

- Only 53 unique labelled profiles support development and evaluation.
- The 0-3 relevance grades contain human judgement uncertainty and no formal inter-rater agreement estimate is available.
- The candidate representation is sparse and relies mainly on job-title text plus basic metadata.
- Lexical retrieval can miss deeper semantic equivalence.
- Repeated experimentation on a small labelled set can overfit model-selection decisions.
- Offline ranking metrics do not establish recruiter productivity or hiring quality.
- Fairness is not guaranteed by omitting protected attributes; text, location, and network-related variables can still act as proxies.

7. Reproduction

The repository is flat. Install requirements.txt, launch Jupyter from the repository root, and run 01_data_representation.ipynb, 02_grouped_cv_model_selection.ipynb, and 03_final_ranking_stage2.ipynb in order. The notebooks read the two CSV files by bare filename and require no helper modules, parent-folder search, build directories, generated result folders, or external figure files.